

Enhancing Security of Grid Control Systems through In-network Collaboration of Programmable Data Planes

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Abstract—The Distributed Network Protocol 3 (DNP3) is widely used in Supervisory Control and Data Acquisition (SCADA) systems for grid monitoring and control, but its limited built-in security leaves many deployments vulnerable to cyberattacks. Programmable data planes such as P4 provide a practical way to move part of DNP3 security logic into the network by enabling protocol-aware packet inspection and in-network mitigation. However, existing P4-based defenses for DNP3 largely rely on standalone switches that make decisions using only local observations, which limits their effectiveness against distributed attacks. In this work, we present a collaborative Multi-P4 framework for detecting and mitigating distributed DNP3 attacks. In the proposed design, each P4 switch monitors local DNP3 traffic, extracts security-relevant features, and exchanges compact metric packets with other switches to build an aggregated view of network behavior. The aggregated information is then used to support system-level detection and coordinated mitigation. We evaluate the framework using two distributed DNP3 attack case studies, cold/warm restart and event buffer flooding. Experimental results show that Multi-P4 detects and mitigates distributed attacks that standalone Single-P4 fails to capture, while improving outstation availability, reducing packet loss, and maintaining low delay for legitimate traffic.

Index Terms—SCADA System, Programmable Networks, P4, DNP3, Network Security

I. INTRODUCTION

The Distributed Network Protocol 3 (DNP3) underpins mission-critical communication in Supervisory Control and Data Acquisition (SCADA) systems, yet its limited built-in security leaves many deployments vulnerable to cyberattacks. In power grid infrastructure, DNP3 is widely used to carry monitoring, status, and control information between control centers and remote field devices [1]. However, many deployments lack strong authentication, integrity protection, and encryption, exposing them to attacks such as false data injection, unauthorized control commands, transport sequence manipulation, cold/warm restart abuse, and event buffer flooding [2]–[6]. These attacks can mislead operators, disrupt measurements, degrade device availability, and even trigger unsafe control actions.

Programmable networking offers a practical way to strengthen DNP3 security inside the network. Unlike traditional switches that mainly forward packets using fixed rules, programmable switches can inspect protocol-specific traffic and apply customized processing in the data plane. P4 (Programming Protocol-independent Packet Processors)

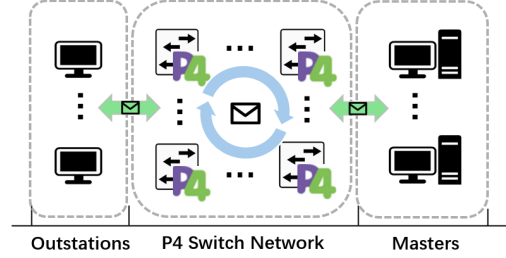


Fig. 1: P4 Collaboration Framework for DNP3 Networks

enables developers to define how packets are parsed, which fields are extracted, and what actions are taken in the switch pipeline [7]. This is especially useful for DNP3, whose well-defined function codes, sequence numbers, addresses, and control commands can be inspected directly in the network. Prior work has shown that P4 switches can support DNP3-aware inspection, filtering, hashing, and mitigation, moving part of DNP3 security logic from external monitoring systems into the network data plane [8].

Despite these benefits, existing P4-based defenses for DNP3 and smart-grid traffic largely follow a standalone model. In this paper, “Single-P4” does not mean that only one P4 switch exists in the network; rather, it means that each switch makes detection and mitigation decisions using only its own local observations. This approach works well when attack evidence is visible at a single observation point, such as a malformed packet or a high-rate flow. However, it is less effective against distributed attacks, where each source may remain below the local detection threshold while the aggregate traffic across multiple paths is sufficient to disrupt the DNP3 system.

To address the limitations of Single-P4 systems, we develop a collaborative Multi-P4 defense for detecting and mitigating distributed DNP3 attacks. Figure 1 illustrates the proposed framework. Each P4 switch monitors local DNP3 traffic, extracts security-relevant features, and generates custom metric packets that summarize local observations over a measurement window. These summaries are exchanged among switches and combined with local observations through register- and table-based data structures to build an aggregated view of network behavior. This collaborative design enables detection of distributed attacks that are not reliably visible from any single switch, supports decisions based on system-level behavior rather than per-

TABLE I: Standalone P4 vs. Collaborative Multi-P4 for DNP3 Security - Attack Characterization and Detection Approach

Attack	Standalone P4	P4 Cooperation	P4 Cooperation Advantage	P4 Cooperation Cost
Substation Circuit Breaker Takeover [2], [3]	Local command allowlist; SBO sequence checking; Command-rate monitoring	Cross-path operation correlation	Split-path cmd injection detection; source/path localization	Per-command state, inter-switch telemetry
Measurement Jitter [4]	Local inter-arrival-time; delay-variation analysis	Cross-switch timing correlation across paths	Low false positives	Time synchronization and timing metadata overhead
Transport Sequence Modification [5]	Local sequence checking; FIR/FIN transition checks	Distributed sequence integrity detection	Detect and localize in-path modification, replay, injection	Data plane stateful registers usage
Cold and Warm Restart [†] [9]	Function-code allowlist; rate-limit checks	Multi-source restart inference	Better source/path attribution	Complexity in state correlation
Event Buffer Flooding [†] [6]	Local poll/event-rate checks	Aggregation of class-poll bursts, unsolicited/event	Detect distributed low-rate flooding	Telemetry volume and aggregation overhead

[†] Cold/warm restart and event buffer flooding are implemented and evaluated in Section V. The remaining attacks represent planned extensions.

switch thresholds, and enables coordinated mitigation by identifying contributing switches, paths, or traffic sources and applying rate limiting or packet dropping near the attack ingress. We evaluate the framework using two distributed DNP3 attack case studies, cold/warm restart and event buffer flooding, and show that Multi-P4 detects and mitigates attacks that standalone Single-P4 fails to capture.

The main contributions of this paper are as follows:

- To the best of our knowledge, this is the first study of collaborative Multi-P4 defense for DNP3-based SCADA networks, and it shows why standalone Single-P4 detection is insufficient for distributed DNP3 attacks.
- We present a qualitative comparison of Single-P4 and Multi-P4 approaches across representative DNP3 attack scenarios, including control-command abuse, timing manipulation, sequence modification, restart attacks, and event buffer flooding.
- We develop a working Multi-P4 collaboration framework that supports metric-packet generation, cross-switch feature sharing, aggregation of local and shared observations, and coordinated mitigation decisions.
- We implement and evaluate two distributed DNP3 attack case studies, cold/warm restart and event buffer flooding attacks, and show that Multi-P4 detects attacks missed by standalone Single-P4 while improving system availability, packet loss, and delay.

II. BACKGROUND AND RELATED WORK

A. DNP3 Security Challenges

Prior work has shown that DNP3 systems are vulnerable to a range of attacks, including control-command manipulation and circuit breaker takeover attacks [2], [3], delay and jitter attacks [4], cold/warm restart abuse [9], and event buffer flooding [6]. These attacks can disrupt communication, trigger unauthorized control actions, degrade monitoring accuracy, or affect the availability and normal operation of field devices.

A key challenge in protecting DNP3 traffic is that some attacks cannot be reliably identified from a single local observation point. When malicious behavior is distributed across multiple paths, switches, or sources, the traffic

observed by one switch may appear benign or remain below a local detection threshold. However, the combined behavior across the network can reveal a coordinated attack pattern. This limitation is especially relevant for attacks such as event buffer flooding and restart abuse, where multiple low-rate malicious sources may jointly create a harmful system-level effect even though each individual source appears locally normal [9].

B. Attack Detection and Mitigation with Programmable Networks

Programmable networking has been explored as a way to improve DNP3 security by enabling network-level detection and mitigation logic to be updated more flexibly than in fixed-function infrastructures. Existing programmable-network defenses for DNP3 generally fall into two categories: SDN-based approaches and P4-based approaches.

1) *SDN-Based Approaches*: In SDN-based systems, a logically centralized controller collects traffic information and dynamically installs forwarding, filtering, or redirection rules after suspicious behavior is detected. This architecture can support intrusion detection and prevention by allowing the network to block, rate-limit, or reroute malicious DNP3 traffic in response to observed threats [10]. The main advantage of SDN is its centralized control and broader network view. However, SDN-based defenses also depend heavily on the controller for decision-making. This dependence can introduce control-plane overhead, increase response latency, and create scalability or reliability concerns in time-sensitive industrial environments.

2) *P4-Based Approaches*: P4 programmable switches provide another approach by moving part of the DNP3 security logic into the data plane. Prior work has shown that P4 switches can support protocol-aware DNP3 parsing and in-network anomaly detection [8]. This enables DNP3-specific features to be extracted and checked directly along the traffic path, allowing some attacks to be detected without relying entirely on a centralized controller. Compared with purely controller-driven SDN approaches, data-plane detection can reduce reaction time because suspicious packets can be identified, marked, dropped, or rate-limited closer to where they are observed.

TABLE II: Comparison of Raw DNP3 Parameters and Derived Features for Single-P4 and Multi-P4 Approaches

Attack	Raw Parameters	Single-P4 Derived Features	Multi-P4 Derived Features
Substation Circuit Breaker Takeover [2], [3]	Control point ID; CROB operation fields; src/dst IP; function codes (Select, Operate, Direct Operate [Ack/No Ack])	Command frequency; source allowlist; select-operate order	Cross-switch select-operate order consistency; cross-switch unauthorized command correlation per endpoint
Measurement Jitter [4]	Timestamp; flow/session ID; packet size	Inter-arrival time	Cross-switch delay variation; path-level timing inconsistency
Transport Sequence Modification [5]	Transport/application sequence numbers; FIR/FIN bits; link-layer addresses	Illegal FIR/FIN combination; sequence jump, rollback, or repeat	Cross-switch FIR/FIN consistency; path-level sequence number inconsistency
Cold and Warm Restart [9]	<code>cold_restart/warm_restart</code> function code; src/dst IP	Frequency of restart commands; unauthorized restart source	Cross-path restart request/response correlation; source/path attribution
Event Buffer Flooding [6]	Function code; timestamp; src/dst IP; packet size	Poll/event-response rate	Cross-switch aggregation of event bursts toward the same outstation

However, existing P4-based defenses for DNP3 and smart-grid traffic mainly focus on detection within a single programmable switch [11]. This design is effective when the relevant attack evidence is visible within one traffic stream or at one observation point. It is less effective for distributed or path-dependent attacks whose indicators are spread across multiple switches, paths, or sources. This limitation motivates a shift from standalone single-P4 monitoring to collaborative multi-P4 monitoring. In a collaborative design, multiple P4 switches can export or exchange compact summaries of locally observed DNP3 features, allowing the network to construct a broader view of DNP3 behavior and detect attack patterns that are not visible to any individual switch alone.

III. STANDALONE P4 VS COLLABORATIVE P4

Tables I and II compare standalone and collaborative P4 defenses in terms of capability, visibility, derived features, and implementation cost. Standalone P4 switches make decisions using only locally observed traffic such as command frequency, inter-arrival time, source authorization, and sequence-number consistency at a single observation point. This approach is effective when the relevant evidence is contained within one traffic stream, such as malformed command or a high-rate flow. However, it can miss distributed attacks in which malicious traffic is divided across multiple sources or paths so that each local observation remains below its detection threshold.

Collaborative Multi-P4 extends this visibility by exchanging compact summaries of locally observed features and correlating them across switches. From the same raw DNP3 fields, it can derive cross-switch features such as SELECT-OPERATE consistency across paths, path-level timing variation, distributed sequence inconsistency, multi-source restart frequency, and the aggregate event-traffic rate toward the same endpoint. These features expose coordinated behavior that is not visible to any individual switch and can improve source or path attribution for mitigation.

This broader visibility introduces communication and coordination costs. Collaborative switches must generate and parse metric packets, align observations within common measurement windows, and maintain state received

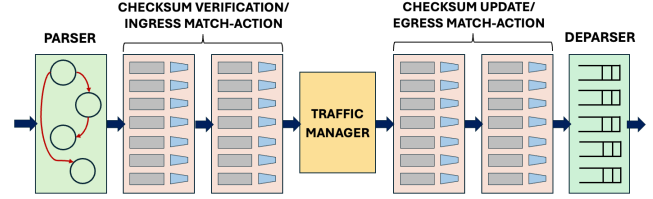


Fig. 2: P4 PISA Architecture

from neighboring switches. Timing-oriented attacks may additionally require synchronization metadata. Because the exchanged messages contain aggregated counters, timestamps, or flow summaries rather than complete packet traces, their bandwidth cost remains substantially smaller than mirroring monitored traffic.

Collaborative detection also consumes additional data-plane resources. Custom headers carry shared summaries, persistent registers maintain local and cross-switch state, and match-action logic identifies metric packets, updates aggregate values, and compares them with mitigation thresholds. The required state depends on the attack: sequence manipulation requires per-flow sequence state across observation points, whereas event buffer flooding requires aggregation across sources targeting the same destination. These costs are absent in standalone P4 systems, but they enable system-level detection, localization, and coordinated mitigation for distributed or path-dependent DNP3 attacks.

IV. DESIGN

A. P4 Architecture

P4 allows creation of a custom packet processing pipeline within a network switch. Figure 2 shows the standard P4 Protocol Independent Switch Architecture (PISA) architecture with programmable parsers and match-action tables (for fast look-ups on parsed fields). P4 programs can run on a variety of targets, including software switches like BMv2 and hardware switches such as Intel’s Tofino ASIC. Depending on the switch target, several other ‘extern’ functions, such as registers, meters, and counters, are available, which aid in stateful packet processing.

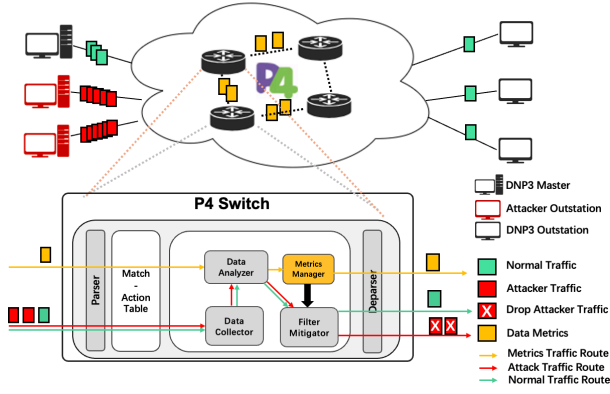


Fig. 3: Proposed P4 Collaborative System Workflow

B. Design Overview

Figure 3 shows our proposed multi-P4 collaboration framework. Legitimate DNP3 packets are sent from masters or outstations to the network, under the surveillance of the multi-P4 structure. Each P4 switch is connected to one or more DNP3 devices and P4 switches. The high-level workflow of the multi-P4 collaboration system is as follows: P4 switches monitor local traffic, extract and share network information, integrate it locally, and forward it to local controllers for policy enforcement. When DNP3 traffic enters the multi-P4 collaboration system, each P4 switch extracts relevant traffic features from its local traffic while preserving normal packet forwarding. The extracted information is shared with other P4 switches and combined with the local traffic information maintained at each switch. This integrated, region-level information is sent to the local P4 controller. Based on this information, the controller issues P4 action configurations to mitigate potential attack traffic, such as adaptive rate limiting.

C. Threat Model

The threat model involves attacker hosts in the DNP3 network with the objective of achieving a denial-of-service, either through an undesired restart of an outstation, or by overwhelming master host buffers. We make the following assumptions on the attacker’s capabilities:

- Full or partial knowledge of the network topology (IP addresses, ports, etc.)
- Ability to spoof DNP3 master or outstation hosts
- Compute power to send DNP3 requests at high rates
- Inability to forge packets used for P4 switch-to-switch communication

We assume trusted P4 switch-to-switch communication. Lightweight integrity and authentication are left as future work.

D. Multi-P4 Communication

1) *Generate and forward custom packets:* The generation of custom headers is achieved by cloning and modifying cloned packets. The header contains the original features of the packets extracted by P4 (such as destination IP) and the processed derived features (such as frequency and throughput). The forwarding of the generated custom

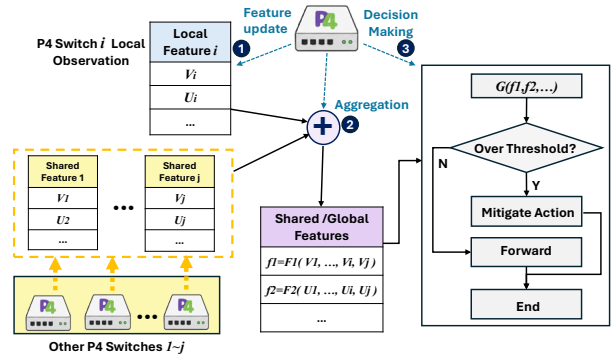


Fig. 4: Inter-switch Aggregation Workflow

packets is customizable, and can be sent to other P4s without affecting the forwarding of the original packets. The features detected by each P4 switch are tracked per time window (preset to 1s in our evaluation), to reduce communication overhead and improve detection stability.

E. Data Structure Integration and Mitigation

Figure 4 illustrates the inter-switch aggregation workflow used in the proposed multi-P4 collaboration scheme. First, each P4 switch i monitors its local traffic and extracts local features, such as V_i , U_i , and other packet-level or flow-level statistics. These features are stored in the local observation register and updated within each measurement time window.

Second, each switch receives shared feature summaries from other P4 switches and integrates them with its own local observations. The integration strategy (F_1 , F_2 in Figure 4) depends on the field type. For traffic-risk rating fields, traffic associated with the same flow ID is aggregated across switches within the same time window to obtain a global view of the flow behavior. In addition, for traffic from different source IP addresses toward the same destination IP address, the system maintains a table-based structure using registers to summarize the cumulative traffic targeting the same destination, enabling detection of distributed attacks that may not exceed the threshold at any single switch. Through this process, homogeneous data structures from multiple independent P4 switches are merged into a global feature representation. Third, the aggregated global features are then used for mitigation decision-making. For simple threshold-based decisions, the mitigation logic can be executed directly in the data plane to provide a fast response. For more complex analysis, such as adaptive rate limiting or advanced traffic classification, the aggregated statistics can be reported to the local controller and processed in the control plane. If the global metric exceeds a predefined threshold, the system triggers mitigation actions such as rate limiting or packet dropping; otherwise, the traffic is forwarded normally.

F. Enhanced Capabilities over Standalone P4 Designs

Compared to existing standalone P4 solutions, our collaborative multi-P4 system introduces several key design optimizations. First, it incorporates a data-sharing component that enables P4 switches to exchange information

with one another. This allows each switch to share its local observations while simultaneously monitoring local traffic. Second, the proposed system includes a dedicated data integration component for processing the shared data. This component performs secondary consolidation to efficiently summarize and maintain region-level network information while reducing memory overhead. Third, the collaborative multi-P4 system leverages the local controller associated with each P4 switch to analyze the integrated network information and issue control actions. By relying on local P4 controllers for decision-making, rather than external centralized third-party controllers, the system can execute mitigation actions with low latency.

G. Communication and Resource Overhead

The Metrics packet used for inter-switch communication occupies a size of 488 bits (61 bytes). Since communication is triggered only when the local observation exceeds the predefined local threshold, each switch sends at most one Metrics packet in each 1-second time window. For the two-switch implementation evaluated in this work, the worst-case collaboration traffic is therefore $2 \times 488 = 976$ bits/s, or 122 bytes/s. Since the DNP3 traffic rates in our experiments are on the order of hundreds of kbps, this additional traffic is less than 1% of a 100 kbps baseline and becomes even smaller as the traffic rate increases.

H. Process Example

Figure 3 describes the workflow of the Multi-P4 security framework, in comparison with the Single-P4 system. Both approaches have P4 pipelines that contain a Data Analyzer, Data Collector, and Filter Mitigator, while the Multi-P4 approach has an additional Metrics Manager module, specifically for inter-switch communication.

DNP3 packets arrive at the P4 network ingress from legitimate Master hosts and attacker hosts. The packets are parsed, checked against match-action tables, and then forwarded to the Data Collector section that extracts the DNP3 command field. The Data Analyzer then statistically calculates the frequency of the command within a preset time window and stores this information locally. The Filter Mitigator finally makes use of this stored information to make drop/forward decisions on the packet, to either forward the packet to its intended destination or drop it.

For the Multi-P4 system, the Data-Analyzer forwards collected statistics to the Metrics-Manager, which formulates the custom Metrics packet containing the flow rate, throughput, and forwards it to the neighboring P4 switch. The neighboring switch parses the metrics packet, extracts the information, and passes it to its own Metrics-Manager, which aggregates local and received statistics and, via its local control plane, updates the Filter-Mitigator to make drop/forward decisions.

V. CASE STUDY

We evaluate the proposed collaborative multi-P4 architecture using two attack scenarios from Table I: Cold/Warm Restart attack and Event Buffer Flooding attack. These two

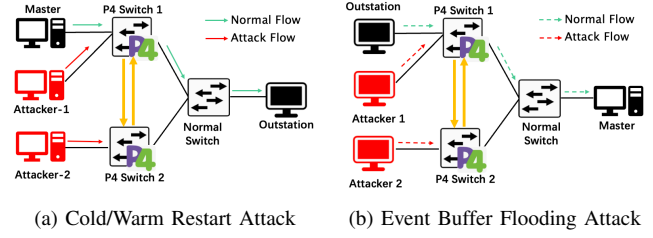


Fig. 5: Evaluation Topologies

attacks are selected because they share a defining property that is central to the motivation of this work: both are distributed low-rate attacks in which each source remains below a locally observable detection threshold, otherwise undetectable by standalone P4 systems. We compare three configurations throughout:

- **Multi-P4 Collaboration System (Multi-P4):** switches exchange aggregated traffic features
- **Standalone P4 System (Single-P4):** switches operate independently using local observations
- **No-P4 System:** baseline without in-network detection

All experiments were conducted using Mininet with the BMv2 software switch on Ubuntu 20.04.6, running in a VirtualBox virtual machine configured with 2 virtual CPUs, 2 GB of RAM, and a 64 GB virtual disk.

A. Cold/Warm Restart Attack

1) *Topology and Attack Pattern:* The evaluation topology is shown in Figure 5a. Two attacker hosts independently send DNP3 restart commands toward the outstation through separate P4 switches. The outstation is considered operational at state 1 and compromised/restarted at state 0. The baseline rate is 2 cmd/s (commands per second). The low-rate attack phase covers rates above baseline up to 10 cmd/s; the high-rate phase is any traffic exceeding 10 cmd/s.

For the Single-P4 system, a local alarm is triggered when the per-switch command rate exceeds 8 cmd/s. For the Multi-P4 system, each switch initiates inter-switch communication when its local command rate reaches 3 cmd/s, and the rate limiting is applied when the aggregated rate across both paths exceeds 7 cmd/s. Two attack modes were evaluated: a distributed low-rate attack, where Attacker 1 and Attacker 2 each send 4 cmd/s (8 cmd/s aggregated, with each source below the local threshold), and a single-source high-frequency attack, where Attacker 1 sends 12 commands/s, and Attacker 2 is inactive.

2) *Results:* Figure 6 shows the outstation state over time for all three systems. Under the single-source high-frequency attack, both Single-P4 and Multi-P4 successfully detect the anomaly and keep the outstation operational. The No-P4 system fails, with the state dropping to state 0.

The distributed low-rate case exposes the key behavior more clearly. With each attacker sending only 4 cmds/s, the per-switch rate never exceeds the Single-P4 alarm threshold (8 cmds/s). The Single-P4 system thus produces no mitigation response, and the outstation is successfully compromised. In contrast, the Multi-P4 system maintains state 1

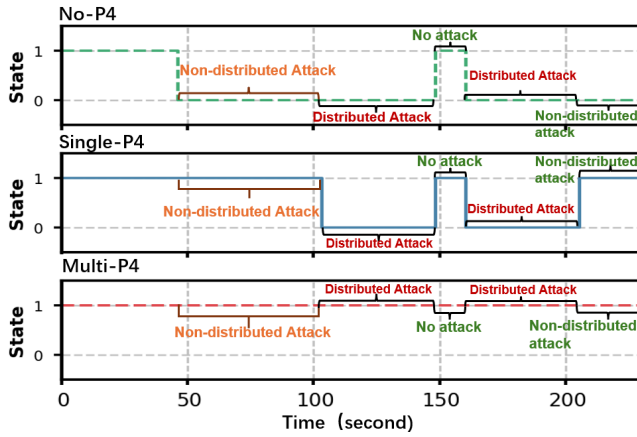


Fig. 6: Outstation State under Cold/Warm Restart Attacks. State 1 indicates the outstation is operational; State 0 indicates that it has been restarted or become unavailable due to a successful attack.

throughout the attack by aggregating restart-command rates across both switches per time window, flags the combined rate as anomalous, enforces rate limiting, and keeps the outstation operational.

B. Event Buffer Flooding Attack

1) *Topology and Attack Pattern:* The evaluation topology is shown in Figure 5b. Two attackers independently send malicious DNP3 traffic toward the master through separate P4 switches. The master begins to experience congestion when the aggregate incoming traffic exceeds approximately 300 kbps, which corresponds to its buffer capacity and processing limit. Legitimate DNP3 traffic is maintained at a constant low rate (10 kbps) and is used to evaluate service degradation under attack.

The detection configurations differ across systems and case studies. In the Single-P4 system, each switch performs detection independently, with the threshold set to 1200 kbps in Fig. 7 and 400 kbps in Fig. 8. In contrast, the Multi-P4 system uses cross-switch coordination. Each switch initiates inter-switch communication when its local traffic exceeds 130 kbps, and mitigation is triggered when the aggregated traffic across both switches exceeds 280 kbps, which closely matches the system congestion point.

The attack follows a two-phase traffic pattern, which is reflected in the time-series plots in Figure 8:

- Low-rate phase (0–25 s): each attacker transmits between 100–300 kbps
- Transition phase (~25–28 s): attack traffic drops briefly to 5 kbps
- High-rate phase (~after 28 s): both attackers increase traffic above 300 kbps

This setup ensures that during the low-rate phase, each individual flow remains below the Single-P4 detection threshold, while the combined traffic already approaches or exceeds the congestion limit.

2) *Results: Average Performance.* Figure 7 shows the average packet loss (left) and packet delay (right) vs. attack rate. Both No-P4 and Single-P4 curves begin to degrade at around 300 kbps, which corresponds to the congestion

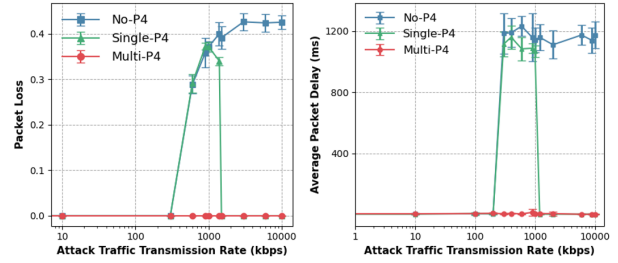


Fig. 7: Average Packet Delay and Loss Ratio

threshold described earlier. This is clearly visible in both subplots: packet loss sharply increases beyond 0.3–0.4, and packet delay rises to ~1000–1200 ms.

This behavior indicates that neither system is able to detect or mitigate the attack at this stage. In particular, Single-P4 behaves almost identically to No-P4 in this region, confirming that local detection is ineffective when traffic is distributed across multiple switches. In contrast, the Multi-P4 curve remains near zero packet loss and maintains low delay (~5 ms) across the entire range. This demonstrates that the collaborative system detects the attack before congestion occurs, based on aggregated traffic information exchanged between switches.

A key observation from this figure is the detection gap: congestion occurs at ~300 kbps (system limit), and the single-P4 detects only at ~1200 kbps (local threshold). This creates a ~900 kbps blind spot, where the system is already overloaded but no mitigation is applied. The Multi-P4 design eliminates this gap by aligning detection with global traffic conditions rather than local thresholds.

Temporal Behavior and Response Dynamics. Figure 8a shows the time-series throughput at the master under both solutions, along with the attack traffic from the two sources. During the low-rate phase (approximately $t = 3\text{--}25$ s), each attacker sends ~150–200 kbps, resulting in ~300–400 kbps total. The Single-P4 system (purple curve) does not react, and the master throughput remains elevated (~350 kbps), indicating that attack traffic is not mitigated. In contrast, the Multi-P4 system (green curve) responds within approximately 1 second, as indicated in the figure annotation, reducing throughput to ~50 kbps after a brief initial spike (~200 kbps). This early response occurs because the Multi-P4 system aggregates traffic across both switches and detects that the combined rate exceeds the global threshold, even though each individual flow remains below the Single-P4 threshold.

During the high-rate phase (around $t = 30\text{--}32$ s), attack traffic increases sharply, producing a large spike. The Single-P4 system finally detects the attack, but only after the master throughput spikes to approximately 1000 kbps, as visible in the purple curve. The Multi-P4 system, however, limits the peak to ~100–150 kbps, significantly suppressing the attack. This difference is due to the persistent mitigation state in Multi-P4. Because the attacker flow ID has already been marked during the low-rate phase, rate limiting remains active as the same flow transitions to

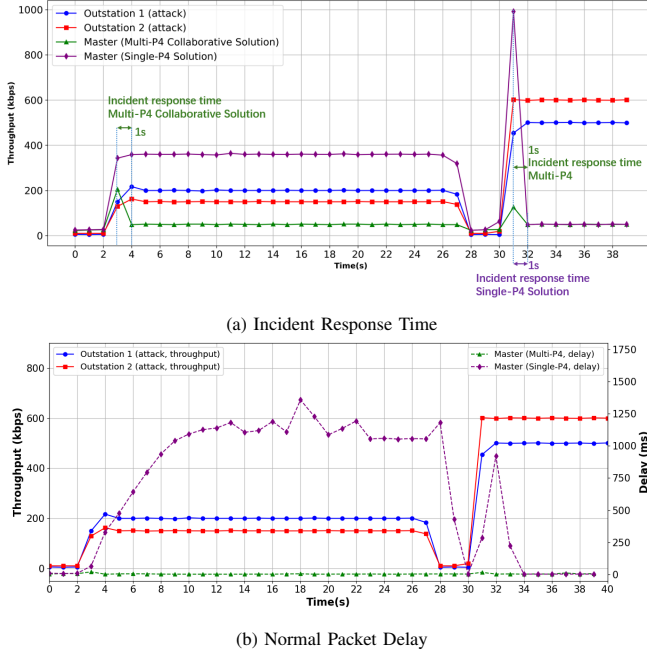


Fig. 8: Comparison of Single-P4 and Multi-P4 Collaborative Solutions under an Event Buffer Flooding Attack

the high-rate phase, keeping forwarded traffic bounded and avoiding the spike seen in the Single-P4 system.

Impact on Legitimate Traffic. Figure 8b shows delay experienced by legitimate DNP3 traffic. Single-P4 introduces severe delay (~ 1000 ms) in both phases due to buffer saturation and lack of early mitigation. The Multi-P4 maintains a stable delay (~ 5 ms) throughout. During the low-rate phase, early mitigation prevents congestion. During the high-rate phase, the existing rate-limiting policy continues to suppress traffic, avoiding delay spikes even before new actions are issued. This measurement represents end-to-end packet delay rather than isolated switch-processing latency. Although switch-processing latency was not isolated in this study, prior P4-based smart-grid evaluations report target-dependent processing times ranging from tens of milliseconds on BMv2 to hundreds of nanoseconds on hardware switches, remaining within the delay requirements considered in those studies [12].

C. Overall Observations

Both case studies validate the core thesis of this work: distributed attacks that are undetectable at any individual switch can be reliably identified through cross-switch aggregation. When viewed in isolation, per-switch traffic appears benign, and only an aggregated view exposes the attack. Our results also show that the Multi-P4 architecture improves both attack sensitivity and system stability without increasing response latency. This advantage is most evident in the low-rate distributed phase, where Single-P4 fails due to limited local visibility, while Multi-P4 succeeds by leveraging cross-switch aggregation.

VI. CONCLUSION

This paper presents a programmable data-plane collaboration architecture for detecting and mitigating distributed

attacks in DNP3-based SCADA networks. By exchanging aggregated traffic features through custom packets, neighboring P4 switches can identify attacks that are difficult for a standalone P4 switch to detect. Experiments on two attack use-cases show improved low-rate attack detection and mitigation. While this work demonstrates the feasibility of collaborative Multi-P4 using two switches, future work will extend this work to larger multi-switch deployments and implement it on physical hardware. These deployments can help evaluate additional attack scenarios such as circuit breaker takeover and transport sequence modification. Scaling to larger deployments will also require investigating how aggregation latency and coordination overhead vary with network topology and the aggregation strategy employed.

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